

1.1 NLP in the Real World

What NLP is, the core tasks, and why human language is a hard input.

1. What NLP is

Natural language processing (NLP) is how we get a computer to work with **human language**: read it, write it, search it, translate it, or answer a question about it.

It sits between linguistics, computer science, and artificial intelligence. The input is messy text (or speech turned into text). The output is a label, a span, a ranking, another sentence, or a number.

This course is **applied**. The goal is to build systems that do a job, not to survey every theory of language.

The main text for the ideas this week is *Practical Natural Language Processing* (Vajjala, Majumder, Gupta, Surana).

2. Tasks you will see all semester

A small set of jobs shows up in almost every project:

Task	What you ask the system to do
Language modeling	Predict the next word from the words so far
Text classification	Put a document into a known bucket (spam, topic, sentiment)
Information extraction	Pull out people, dates, events, or other fields
Information retrieval	Find documents that match a query
Conversational agent	Keep a dialogue going in ordinary language
Text summarization	Shorten a document and keep the point
Question answering	Answer a question written in English
Machine translation	Rewrite the text in another language
Topic modeling	Find themes in a large collection of documents

Later weeks pick these up one at a time. Week 1 is the map.



Text
Classification



Information
Extraction



Conversational
Agent



Information
Retrieval



Question
Answering Systems



Spam
Classification



Calendar Event
Extraction



Personal
Assistants



Search
Engines

JEOPARDY!

Jeopardy!



Social Media
Analysis



Retail Catalog
Extraction



Health Records
Analysis



Financial
Analysis



Legal Entity
Extraction

3. Why the job is hard

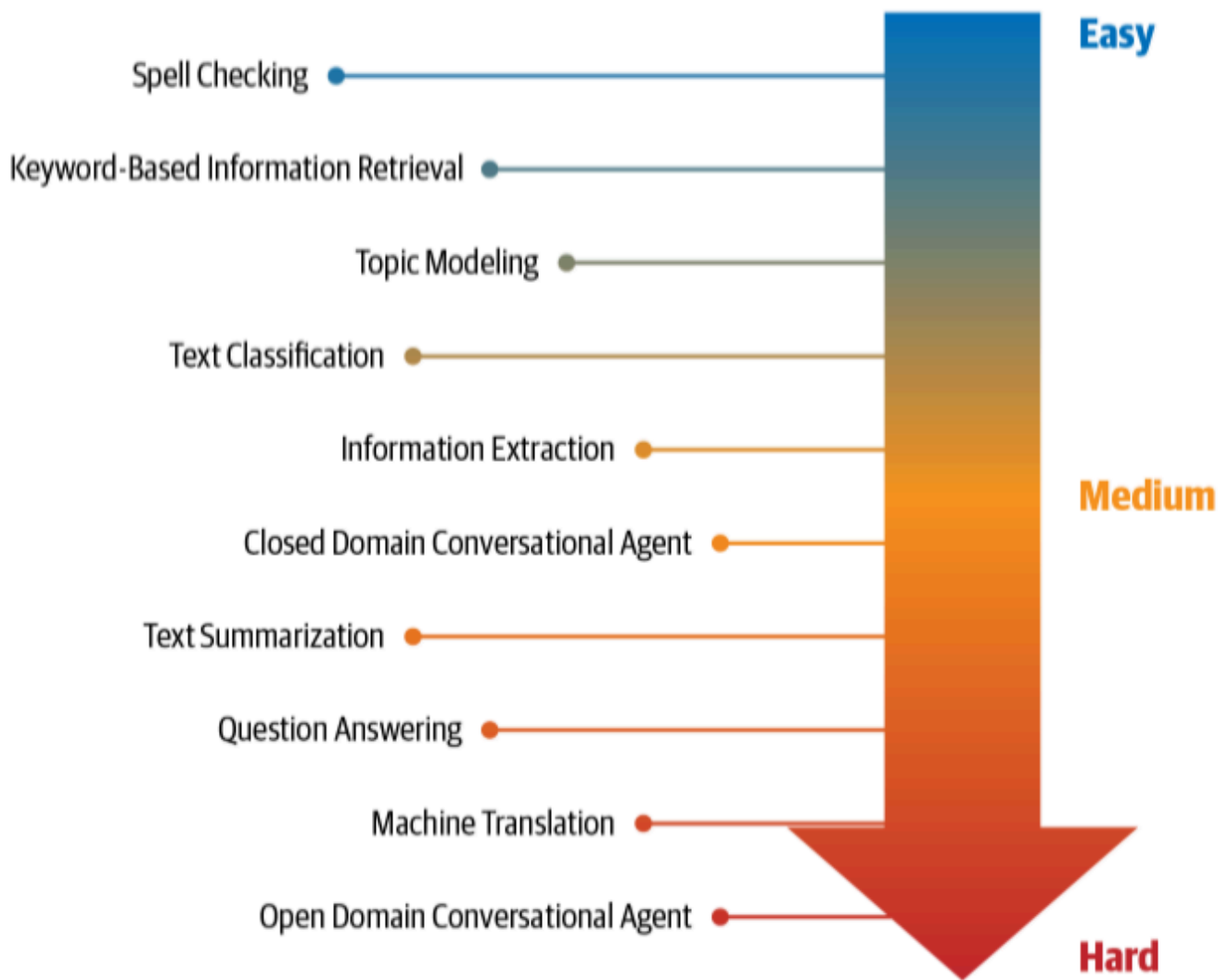
Human language is not a clean table.

Ambiguity. The same string can mean more than one thing. *I saw her duck* is a bird or a movement. A model has to use context.

Common knowledge. Speakers leave facts unsaid. "The restaurant was packed, so we left" assumes you know what a packed restaurant is like. The text does not spell that out.

Creativity. Dialects, slang, sarcasm, and new coinages break rules you wrote last month.

Diversity across languages. There is no one-to-one map between vocabularies. A pipeline that works for English does not port by renaming a folder.



The man couldn't lift his son because he was so **weak**. —○ Who was weak?

The man couldn't lift his son because he was so **heavy**. —○ Who was heavy?

Mary and Sue are **sisters**.
Mary and Sue are **mothers**. } —○ How are Mary and Sue related?

Joan made sure to thank Susan for all the help she had **received**. —○ Who had received help?

Joan made sure to thank Susan for all the help she had **given**. —○ Who had given help?

John **promised** Bill to leave, so an hour later he left.
John **ordered** Bill to leave, so an hour later he left. } —○ Who left an hour later?

4. What this week is for

By the end of Week 1 you should be able to:

- name the core NLP tasks in the table above
- say why a rule-only system struggles
- write a short Python script that uses lists, dictionaries, functions, and files

Note **1.2** is how machine learning and deep learning sit next to those tasks. Note **1.3** is the Python you need for Lab 1.

5. Practice

1. Pick a product you use (search, mail, maps, a chatbot). Name two NLP tasks from the table that it is doing.
2. Write one English sentence that is ambiguous. Say the two readings.
3. Why would a perfectly good English spam filter do poorly on another language even if you translate the training labels?